

Software Engineering

Lab 3 – Architecture Diagram

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Justification:

Architecture Selection: We chose **Layered Architecture** for the **Pet Adoption & Shelter Management System**.

Architectural Choice: The system is organised into three layers. The Presentation layer holds the Adoption Applicant Portal and the Shelter Staff Portal. The Business Logic layer holds Adoption Management, Animal & Medical Records, and Foster Placement Management. The Data layer holds the Shelter Database. Each layer calls only the layer directly below it, through provided and required interfaces such as IApplication, IPetSearch and IAnimalData.

Reason 1: The workflow maps onto layers. Applicants and shelter staff use different screens but must obey the same rules. For example, an animal under medical quarantine must never be marked adopted. Placing this rule once in the business layer (the IAnimalStatus interface) makes both portals enforce it identically. Neither portal needs its own copy, so the two cannot disagree.

Reason 2: The scale and data model favour a simple structure. Animals, adoption applications and foster placements all link through the same animal record, and one shelter serves a modest number of users. A single shared database with three business modules fits this well. Microservices would add network calls, separate deployments and distributed data-consistency problems that a single shelter does not need. Plain client-server would mix interface code with business rules, making changes harder.

Security Advantage: The portals never touch the database. All access to sensitive data, such as applicants' questionnaires and animals' medical histories, passes through named business-layer interfaces. Input validation, applicant and staff role checks, and parameterised SQL queries are therefore applied in one place, and the attack surface is limited to those interfaces.

Performance Benefit: Pet search must support multi-attribute filtering (species, age, breed, good with kids) with a page response under 200 ms. The search sits behind its own interface (IPetSearch) on the Animal & Medical Records component. It can therefore be tuned with database indexes on the filter columns, result pagination and caching without changing the application or foster modules.